

Answer Key & Marking Scheme

Subject: Science | **Test:** Test 4 (Day 28, 6 June 2026) — 2-Week Cumulative
Maximum Marks: 80

Syllabus: Chs 7–12: How do Organisms Reproduce? • Heredity • Light – Reflection and Refraction • The Human Eye and the Colourful World • Electricity • Magnetic Effects of Electric Current

SECTION A — Answer Key (1 mark each)

- Q1. (a) Plasmodium.
- Q2. (b) stigma down the style to the ovary.
- Q3. (a) ovary — hormones from the ovary regulate the cycle.
- Q4. (c) A or B — $(I \times i)$ and $(I \times i) \Rightarrow Ii$ or Ii .
- Q5. (c) 3 : 1 (phenotypic).
- Q6. (b) on the opposite side of the object — per sign convention.
- Q7. (b) C (2F).
- Q8. (b) concave lens.
- Q9. (c) adjust focal length of the lens.
- Q10. (c) scattering.
- Q11. (c) ampere.
- Q12. (a) directly proportional to length.
- Q13. (b) $2\ \Omega$ — $1/R = 1/4 + 1/6 + 1/12 = 3/12 + 2/12 + 1/12 = 6/12 = 1/2 \Rightarrow R = 2$.
- Q14. (c) direction of magnetic field around a current carrying conductor.
- Q15. (b) a bar magnet.
- Q16. (a) 50 Hz.
- Q17. Magnification = ratio of image height to object height (or v/u with sign convention for lenses).
- Q18. Ohm's law: At constant temperature, the current through a conductor is directly proportional to the potential difference across its ends, $V = IR$.
- Q19. Tungsten has very high melting point ($\sim 3380^\circ\text{C}$) and high resistivity, so it can glow at high temperature without melting.
- Q20. An allele is one of the alternative forms of a gene located at the same position on homologous chromosomes (e.g. T and t for height in pea plants).

SECTION B — Answer Key (2 marks each)

- Q21. **Fragmentation:** a multicellular organism breaks into smaller fragments and each grows into a new individual; example — Spirogyra. **Regeneration:** a body part broken/cut off can grow back into a complete organism, but only specific specialised cells in the original body can do so; example — Planaria, Hydra. (2)
- Q22. Sexual reproduction involves combination of genetic material from two parents through meiosis (independent assortment, crossing over) and random fertilisation. This produces offspring with genetic variation. Asexual reproduction involves a single parent and produces (almost) genetically identical offspring; variation occurs only through chance mutations. (2)
- Q23. Power of a lens $P = 1/f$ (in metres), measured in dioptres (D). For $f = 25\text{ cm} = 0.25\text{ m}$, $P = 1/0.25 = +4\text{ D}$. (2)
- Q24. At sunset, sunlight has to pass through a thicker layer of atmosphere. Most of the shorter (blue) wavelengths get scattered out, leaving the longer (red and orange) wavelengths to reach the observer — making the sky and sun appear red. (2)
- Q25. A fuse is a thin wire of low melting point connected in series with the live wire. When current exceeds a safe value (short circuit/overload), the fuse wire heats up rapidly and melts, breaking the circuit and protecting appliances and wiring from damage. (2)

Q26. $R = V/I = 6/0.5 = 12 \, \Omega$. Time $t = 5 \text{ min} = 300 \text{ s}$. Heat $H = I^2Rt = 0.25 \times 12 \times 300 = \mathbf{900 \text{ J}}$. (2)

SECTION C — Answer Key (3 marks each)

Q27. When a sperm fertilises the egg in the fallopian tube, the resulting zygote begins to divide and forms an embryo. The embryo travels down the tube to the uterus and implants in the uterine lining. The placenta develops, providing the embryo with oxygen and nutrients from the mother's blood and removing wastes. The embryo grows over ~9 months into a foetus, ready for birth. (3)

Q28. Cross: $TT \times tt$. **F₁** Punnett square: T over t (mother) $\times$ T or t (father). All **F₁** offspring are **Tt** (tall, since T is dominant). On self-pollination of **F₁**: **F₂** from $Tt \times Tt$ gives $TT : Tt : Tt : tt = 1 : 2 : 1$ (genotype) and 3 tall : 1 dwarf (phenotype). (3)

Q29. (i) Object beyond 2F: image between F and 2F on opposite side, real, inverted, diminished. (ii) Object at 2F: image at 2F on opposite side, real, inverted, same size. (iii) Object between F and 2F: image beyond 2F on opposite side, real, inverted, magnified. Each ray diagram should show two rays — one parallel to axis refracting through F, one through optical centre. (3)

Q30. Myopia (short-sightedness): the person can see near objects clearly but distant objects appear blurred. The image is formed in front of the retina. **Correction:** a concave (diverging) lens of suitable power, which diverges incoming parallel rays so the eye focuses them on the retina. **Hypermetropia (long-sightedness):** the person can see distant objects clearly but near objects appear blurred. The image is formed behind the retina. **Correction:** a convex (converging) lens of suitable power. (3)

Q31. (i) Series: $R = 2 + 3 + 6 = \mathbf{11 \, \Omega}$. (ii) Parallel: $1/R = 1/2 + 1/3 + 1/6 = 3/6 + 2/6 + 1/6 = 6/6 = 1 \Rightarrow R = \mathbf{1 \, \Omega}$. (3)

Q32. An electric motor converts electrical energy into mechanical (rotational) energy. **Construction:** a coil ABCD placed between the poles of a magnet, connected to a battery via brushes **B₁**, **B₂** and a split-ring commutator. **Working:** When current flows through the coil, the magnetic field exerts equal and opposite forces on AB and CD (Fleming's left-hand rule), creating a couple. The coil rotates. The split-ring commutator reverses the current after every half rotation, ensuring the coil continues rotating in the same direction. (3)

Q33. Electromagnetic induction: the production of an electric current in a closed circuit due to a change in the magnetic flux through it. **Faraday's law:** the induced EMF is directly proportional to the rate of change of magnetic flux. **Application:** the electric generator (dynamo) — uses a rotating coil in a magnetic field to produce electricity in power plants. (3)

SECTION D — Answer Key (5 marks each)

Q34. (a) Diagram with labels: cornea, aqueous humour, iris, pupil, ciliary muscles, lens, vitreous humour, retina, optic nerve, blind spot, yellow spot. (3) (b) Light enters through the cornea, passes through the pupil (size controlled by iris), is focused by the convex lens, travels through the vitreous humour and forms an inverted, real image on the retina. Photoreceptor cells (rods and cones) on the retina convert the image into nerve impulses, which are transmitted to the brain via the optic nerve where the image is interpreted (and turned upright). (2)

Q35. (a) (i) Incident ray, refracted ray and normal lie in the same plane. (ii) $\sin i / \sin r = \text{constant}$ (Snell's law), equal to the refractive index of the second medium relative to the first. (1) (b) $u = -30 \text{ cm}$, $f = +20 \text{ cm}$. Lens formula: $1/v - 1/u = 1/f \Rightarrow 1/v = 1/20 + 1/(-30) = 3/60 - 2/60 = 1/60 \Rightarrow v = +60 \text{ cm}$. $m = v/u = 60/(-30) = -2$. Image height $= -2 \times 5 = -10 \text{ cm}$. **Image is at 60 cm on opposite side, real, inverted, magnified (twice as large).** (4)

Q36. (a) **Ohm's law:** at constant temperature, the current I through a conductor is directly proportional to the potential difference V across its ends: $V \propto I \Rightarrow V = IR$, where R (the constant of proportionality) is the resistance of the conductor. V — volts, I — amperes, R — ohms. (2) (b) For two equal resistors R : **Series:** $R_s = 2R$; current $I_s = V/(2R)$; heating power $P_s = V^2/(2R)$. **Parallel:** $R_p = R/2$; current $I_p = 2V/R$; power $P_p = V^2/(R/2) = 2V^2/R$. So **$P_p / P_s = 4$** . Hence the parallel combination dissipates four times more heat than the series combination, for the same applied voltage. (3)

SECTION E — Answer Key (4 marks each)

Q37. (i) Hypermetropia (long-sightedness / far-sightedness). (1) (ii) The eye lens has too little converging power, or the eyeball is too short — so the image of a near object is formed behind the retina. (1) (iii) A convex (converging) lens placed before the eye converges the rays so that the final image falls on the retina. Ray diagram: parallel rays from a near object converge through the spectacle convex lens and then converge at the retina with the eye lens. (2)

Q38. (i) $R = V^2/P$. $R_{\blacksquare}$ (100 W bulb) = $220^2/100 = 484 \Omega$; $R_{\blacksquare}$ (60 W bulb) = $220^2/60 \approx 807 \Omega$. (1) (ii) In **series**, both bulbs carry the same current. Power dissipated $P = I^2R$; the bulb with higher R (the 60 W) dissipates more power and glows brighter. (Counter-intuitively the 'lower wattage' bulb is brighter in series.) (2) (iii) In **parallel**, both bulbs are at 220 V; each dissipates its rated power. The 100 W bulb glows brighter. (1)

Q39. (i) The father's chromosome contribution (X or Y) determines the sex of the child. (1) (ii) Punnett square: father (X, Y) $\times$ mother (X, X) $\rightarrow$ XX, XY, XX, XY. So 50% daughters (XX), 50% sons (XY). (2) (iii) Probability of male child = $1/2 = 50\%$. (1)

— End of Answer Key —